1 Tensors

Tensors are a fundamental concept in mathematics and physics, generalizing scalars, vectors, and matrices to higher dimensions. They provide a powerful language for expressing physical laws and transformations in a coordinate-independent manner. Tensors play a crucial role in many branches of science and engineering, particularly in fields like continuum mechanics, general relativity, and material science, where complex relationships need to be described in a compact and general form.

A tensor can be thought of as a multi-dimensional array that can be indexed by multiple indices, and its components can vary depending on the choice of coordinate system. In this sense, scalars are considered as tensors of order zero, vectors as tensors of order one, and matrices as tensors of order two. Tensors of higher order are also common in physics and engineering, especially in fields like solid mechanics, fluid dynamics, and electromagnetism.

The key property that defines a tensor is its transformation rule under a change of coordinates. When the coordinate system is changed, the components of a tensor transform in a way that ensures the underlying physical quantity remains unchanged. This makes tensors particularly valuable for describing physical phenomena in a way that is independent of the observer's perspective.

1.1 Notation

In textbooks, scalars variables are usually typed in italics, vectors in bold, and higher order tensors in bold and uppercase.

Scalars: Scalars are variables that have only magnitude. They are represented by a single number. For example, the mass of a body m, the temperature of a fluid T, the pressure of a gas p, the density of a material ρ, etc.

Vectors: Vectors are variables that have both magnitude and direction where the latter is composed by two or more components. They are frequently represented by arrows when drawn. Examples of vectors are, the force acting on a body $\boldsymbol{F}$, the velocity of a particle $\boldsymbol{v}$, the displacement on a particular point of a body $\boldsymbol{u}$, etc.

Matrices: They are rectangular arrays of numbers. For example, the rotation matrix $\boldsymbol{R}$, the stiffness matrix $\boldsymbol{K}$, etc.

Tensors: Tensors are a generalization of arrays in higher dimensions. For example, the stress tensor $\boldsymbol{\sigma}$, the strain tensor $\boldsymbol{\varepsilon}$, the elastic consitutive tensor $\boldsymbol{D}$, etc.

The number of components of a tensor depends on the number of dimensions of the space in which it is defined. For example, in 3D space, the stress and strain tensors have 9 components each, while in 2D space, they have 4 components each. Thus, in 3D space we have:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix} \tag{1.1}$$

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix} \tag{1.2}$$

And in 2D space we have

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{11} & \sigma_{12} \\ \sigma_{21} & \sigma_{22} \end{pmatrix} \tag{1.3}$$

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} \\ \varepsilon_{21} & \varepsilon_{22} \end{pmatrix} \tag{1.4}$$

Tensor operations can be described using two different notations: abstract (or direct) notation and index notation. In the abstract notation, tensors are represented using symbols and operators, while in the index notation, the components of tensors are represented using subscripts and, in some cases, superscripts.

For example, given a vector $\boldsymbol{v}$ and a second-order tensor $\boldsymbol{A}$, the matrix-vector product is written in abstract notation as

$$\boldsymbol{c} = \boldsymbol{A}\boldsymbol{v} \tag{1.5}$$

while in index notation, the components of the resulting vector $\boldsymbol{c}$ are written as

$$c_i = \sum_j A_{ij} v_j \tag{1.6}$$

where i and j are the indices of the components.

1.2 Index notation

The components of tensors are usually represented using index notation by using subscripts and superscripts. For example, the components of a vector $\boldsymbol{v}$ can be represented as v_i, where i is the index of the component. The components of a second-order tensor $\boldsymbol{\sigma}$ can be represented as σ_{ij}, where i and j are the indices of the components.

For example, given the tensors

$$\boldsymbol{a} = \begin{pmatrix} 2 \\ 7 \\ 4 \end{pmatrix} \text{ and } \boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{pmatrix} \tag{1.7}$$

we have $a_2 = 7$, $A_{21} = 4$, $A_{32} = 8$, $A_{33} = 9$, etc.

Now, considering two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$

$$\boldsymbol{a} = \begin{pmatrix} a_1 \\ a_2 \\ a_3 \end{pmatrix} \text{ and } \boldsymbol{b} = \begin{pmatrix} b_1 \\ b_2 \\ b_3 \end{pmatrix}, \tag{1.8}$$

the outer product of these vectors is defined as $\boldsymbol{C} = \boldsymbol{a} \otimes \boldsymbol{b} = \boldsymbol{a}\boldsymbol{b}^T$ and written in full as

$$\boldsymbol{C} = \begin{pmatrix} a_1 b_1 & a_1 b_2 & a_1 b_3 \\ a_2 b_1 & a_2 b_2 & a_2 b_3 \\ a_3 b_1 & a_3 b_2 & a_3 b_3 \end{pmatrix} \tag{1.9}$$

where the components are given by $C_{ij} = a_i b_j$.

The index notation is also used to represent derivatives in higher dimensions. For instance, given a vector field $\boldsymbol{f} = (f_1, f_2, f_3)$, we may write its derivatives with respect to the Cartesian coordinates $\boldsymbol{x} = (x_1, x_2, x_3)$ as the second order tensor

$$\nabla \boldsymbol{f} = \frac{\partial \boldsymbol{f}}{\partial \boldsymbol{x}} = \begin{pmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \frac{\partial f_1}{\partial x_3} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_3}{\partial x_1} & \frac{\partial f_3}{\partial x_2} & \frac{\partial f_3}{\partial x_3} \end{pmatrix} \tag{1.10}$$

Each component of $\nabla \boldsymbol{f}$ can be expressed as

$$f_{i,j} = \frac{\partial f_i}{\partial x_j} \tag{1.11}$$

where the first index i refers to the component of the vector field and the second index j refers to the variable with respect to which the derivative is taken.

As another example, given a scalar function $f(x_1, x_2, x_3)$, the gradient of f is represented by

$$\nabla f = \frac{\partial f}{\partial \boldsymbol{x}} = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{pmatrix} \tag{1.12}$$

where each component is expressed as

$$f_{,i} = \frac{\partial f}{\partial x_i} \tag{1.13}$$

1.3 Summation convention

The summation convention, also called Einstein summation convention, is a notational convention that implies summation over repeated indices. For example, the dot product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ in 3D space can be written as $a_i b_i$, where the index i is repeated and the following sum is implied:

$$\boldsymbol{a} \cdot \boldsymbol{b} = \sum_{i=1}^{3} a_i b_i = a_i b_i = a_1 b_1 + a_2 b_2 + a_3 b_3 \tag{1.14}$$

Note the the summation symbol $\sum$ is ommited for implicit summation over repeated indices. Repeated indices are called dummy indices while non repeated indices are called free indices. For example, the resulting components from the matrix vector product $\boldsymbol{c} = \boldsymbol{A}\boldsymbol{b}$ can be written as $c_i = A_{ij} b_j$ where i is a free index and j a dummy index. Considering 3D space, we have

$$c_i = \sum_{j=1}^{3} A_{ij} b_j = A_{ij} b_j = A_{i1} b_1 + A_{i2} b_2 + A_{i3} b_3 \tag{1.15}$$

The expression C_i with one free index indicates that the result of the product is a order-1 tensor, a vector.

The norm of a vector $\boldsymbol{a}$ can be written as

$$\|\boldsymbol{a}\| = \sqrt{\sum_{i=1}^{3} a_i^2} = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}} = \sqrt{a_1^2 + a_2^2 + a_3^2} = \sqrt{a_i a_i} \tag{1.16}$$

Given a second-order tensor $\boldsymbol{A}$, the sum of the terms from the diagonal, also know as the trace of the tensor, can be written as

$$\sum_{i=1}^{3} A_{ii} = A_{ii} = A_{11} + A_{22} + A_{33} \tag{1.17}$$

Given a second-order tensors $\boldsymbol{A}$ and a vector $\boldsymbol{b}$, the components from the matricial product $\boldsymbol{c} = \boldsymbol{A}\boldsymbol{b}$ are given by

$$C_i = \sum_{j=1}^{3} A_{ij}b_j = A_{ij}b_j = A_{i1}b_1 + A_{i2}b_2 + A_{i3}b_3 \tag{1.18}$$

Given two second-order tensors $\boldsymbol{A}$ and $\boldsymbol{B}$, the components from the matricial product $\boldsymbol{C} = \boldsymbol{A}\boldsymbol{B}$ are given by

$$C_{ij} = \sum_{k=1}^{3} A_{ik}B_{kj} = A_{ik}B_{kj} = A_{i1}B_{1j} + A_{i2}B_{2j} + A_{i3}B_{3j} \tag{1.19}$$

The components of transpose matrix $\boldsymbol{B}^T$ are given by $B^T_{jk} = B_{kj}$, then the components from the product $\boldsymbol{A}\boldsymbol{B}^T$ are

$$C_{ij} = \sum_{k=1}^{3} A_{ik}B_{jk} = A_{ik}B_{jk} = A_{i1}B_{j1} + A_{i2}B_{j2} + A_{i3}B_{j3} \tag{1.20}$$

Given a vector field $\boldsymbol{f}$, the divergence of $\boldsymbol{f}$ is represented as

$$\nabla \cdot \boldsymbol{f} = \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} + \frac{\partial f_3}{\partial x_3} = f_{1,1} + f_{2,2} + f_{3,3} = f_{i,i} \tag{1.21}$$

Note 1.1: In the summation convetion, an index cannot appear more than twice within the same term. If an expression require that an index to appear more than twice, the summation must be performed explicitly. For example:

$$\boldsymbol{A} = \sum_{i=1}^{3} \lambda_i \boldsymbol{v}_i \otimes \boldsymbol{v}_i$$

Additionally, if an index appears twice but is not intended to be summed, an explicit indication is needed to avoid ambiguity. For example:

$$\boldsymbol{P}_i = \boldsymbol{v}_i \otimes \boldsymbol{v}_i \quad \text{(no summation on } i\text{)}$$

Here, the note "(no summation on i)" clarifies that the index i is not summed over.

Example 1.1: Considering 3D space, expand the expression αA_{jj}.

Solution:

For 3D space, $j = 1, 2, 3$, then

$$\begin{aligned} \alpha A_{jj} &= \alpha \sum_{j=1}^{3} A_{jj} \\ &= \alpha A_{11} + \alpha A_{22} + \alpha A_{33} \end{aligned}$$

Example 1.2: Considering 3D space, find all the components of the matrix-vector product $\boldsymbol{Ab}$.

Solution:

Assuming $\boldsymbol{c} = \boldsymbol{Ab}$ and following the index notation, each component of the product have the form

$$c_i = A_{ij} b_j$$

then, applying the summation convention on index $j = 1, 2, 3$ we have

$$c_i = A_{i1} b_1 + A_{i2} b_2 + A_{i3} b_3$$

Finally, for $i = 1, 2, 3$, the components of the resulting vector $\boldsymbol{c}$ are

$$\begin{aligned} c_1 &= A_{11} b_1 + A_{12} b_2 + A_{13} b_3 \\ c_2 &= A_{21} b_1 + A_{22} b_2 + A_{23} b_3 \\ c_3 &= A_{31} b_1 + A_{32} b_2 + A_{33} b_3 \end{aligned}$$

Example 1.3: Find the resulting expression for $(\boldsymbol{Ab}) \cdot \boldsymbol{c}$.

Solution:

The resulting expression is a scalar quantity

$$\alpha = (A_{ij} b_j) c_i$$

Applying the summation on index j we have

$$\alpha = (A_{i1} b_1 + A_{i2} b_2 + A_{i3} b_3) c_i$$

Finally, applying the summation on index i we have

$$\begin{aligned} \alpha = {} & A_{11} b_1 c_1 + A_{12} b_2 c_1 + A_{13} b_3 c_1 + \\ & A_{21} b_1 c_2 + A_{22} b_2 c_2 + A_{23} b_3 c_2 + \\ & A_{31} b_1 c_3 + A_{32} b_2 c_3 + A_{33} b_3 c_3 \end{aligned}$$

Exercises 1.1:

1. Given the tensors

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix} \quad \text{and} \quad \boldsymbol{b} = \begin{pmatrix} 1 \\ 0 \\ 2 \end{pmatrix}$$

calculate the values of

a) A_{ii}
b) $A_{ij} b_j$
c) $A_{ji} b_j$
d) $b_i b_j$
e) $A_{ij} A_{ji}$

2. Given the expressions in index notation, write the corresponding expressions in abstract tensor notation.

a) a_ib_i
b) $a_ib_kc_k$
c) $A_{ik}B_{kj}$
d) $A_{ki}B_{kl}A_{lj}$
e) $a_iA_{jk}b_k$
f) $a_iA_{ik}b_k$

1.4 Kronecker delta

The Kronecker delta is a function of two variables that is used to represent the components of the identity matrix. It is defined as:

$$\delta_{ij} = \begin{cases} 1 \text{ if } i = j \\ 0 \text{ if } i \neq j \end{cases} \tag{1.22}$$

The second-order identity tensor can be expressed in terms of the Kronecker delta as

$$\boldsymbol{I} = \begin{pmatrix} \delta_{11} & \delta_{12} & \delta_{13} \\ \delta_{21} & \delta_{22} & \delta_{23} \\ \delta_{31} & \delta_{32} & \delta_{33} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \tag{1.23}$$

The Kronecker delta has the following properties:

$$\begin{aligned} \delta_{ij} &= \delta_{ji} \quad \text{symmetry} \\ \delta_{ii} &= 3 \qquad\quad \text{trace} \\ \delta_{ij}\delta_{jk} &= \delta_{ik} \text{ contraction} \end{aligned} \tag{1.24}$$

When used in multiplications, the indices from the Kronecker delta can be used to perform an index substitution simplifying the expression. For instance, given a second-order tensor $\boldsymbol{A}$ and a vector $\boldsymbol{b}$ we have

$$\begin{aligned} \delta_{ij}b_j &= b_i \\ \delta_{ij}A_{jk} &= A_{ik} \\ \delta_{ij}A_{kj} &= A_{ki} \text{ transpose} \end{aligned} \tag{1.25}$$

Example 1.4: Simplify the expression $\delta_{ij}a_j$.

Solution:

Expanding the expression for index j we have

$$\delta_{ij}a_j = \delta_{i1}a_1 + \delta_{i2}a_2 + \delta_{i3}a_3$$

Now, for each value of i we have

$$\begin{aligned} \delta_{1j}a_j &= \delta_{11}a_1 + \delta_{12}a_2 + \delta_{13}a_3 \\ \delta_{2j}a_j &= \delta_{21}a_1 + \delta_{22}a_2 + \delta_{23}a_3 \\ \delta_{3j}a_j &= \delta_{31}a_1 + \delta_{32}a_2 + \delta_{33}a_3 \end{aligned}$$

Applying the Kronecker delta definition

$$\begin{aligned} \delta_{1j}a_j &= a_1 \\ \delta_{2j}a_j &= a_2 \\ \delta_{3j}a_j &= a_3 \end{aligned}$$

Finally, we end up with 3 components of the vector $\boldsymbol{a}$, then the expression is simplified as

$$\delta_{ij}a_j = a_i$$

This result represents the contraction of the Kronecker delta with vector $\boldsymbol{a}$.

Example 1.5: Simplify the expression $\delta_{ij}\delta_{ji}$.

Solution:

Applying the contraction property of the Kronecker delta we have

$$\delta_{ij}\delta_{ji} = \delta_{ii} = 3$$

Example 1.6: Find the trace of the product of two second-order tensors $\boldsymbol{A}$ and $\boldsymbol{B}$.

Solution:

The product of the two tensors is given by $\boldsymbol{C} = \boldsymbol{AB}$ where the components are

$$C_{ij} = A_{ik}B_{kj}$$

The trace, $\alpha = \text{trace}(\boldsymbol{AB})$, can be obtained by contracting the product $\boldsymbol{AB}$ with the aid of the Kronecker delta as

$$\begin{aligned}\alpha &= A_{ik}B_{kj}\delta_{ij} \\ &= A_{ik}B_{ki} \\ &= A_{i1}B_{1i} + A_{i2}B_{2i} + A_{i3}B_{3i} \\ &= A_{11}B_{11} + A_{12}B_{21} + A_{13}B_{31} + \\ &\quad\; A_{21}B_{12} + A_{22}B_{22} + A_{23}B_{32} + \\ &\quad\; A_{31}B_{13} + A_{32}B_{23} + A_{33}B_{33}\end{aligned}$$

Example 1.7: Find a expression for the derivative of a vector $\boldsymbol{v}$ with respect to itself.

Solution:

The derivative of a vector $\boldsymbol{v}$ with respect to itself can be expressed as

$$\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{v}} = \begin{pmatrix} \frac{\partial v_1}{\partial v_1} & \frac{\partial v_1}{\partial v_2} & \frac{\partial v_1}{\partial v_3} \\ \frac{\partial v_2}{\partial v_1} & \frac{\partial v_2}{\partial v_2} & \frac{\partial v_2}{\partial v_3} \\ \frac{\partial v_3}{\partial v_1} & \frac{\partial v_3}{\partial v_2} & \frac{\partial v_3}{\partial v_3} \end{pmatrix} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Using index notation we can express the components as

$$\frac{\partial v_i}{\partial v_j} = \delta_{ij}$$

Example 1.8: Considering that the components of $\boldsymbol{A}$ are constants, simplify the expression $\frac{\partial}{\partial x_k}(A_{ij}x_ix_j)$

Solution:

Performing the derivative with respect to x_k we have

$$\begin{aligned}
\frac{\partial}{\partial x_k}(A_{ij}x_ix_j) &= A_{ij}\frac{\partial}{\partial x_k}(x_ix_j) \\
&= A_{ij}\left(\frac{\partial x_i}{\partial x_k}x_j + x_i\frac{\partial x_j}{\partial x_k}\right) \\
&= A_{ij}(\delta_{ik}x_j + x_i\delta_{jk}) \\
&= A_{ij}\delta_{ik}x_j + A_{ij}x_i\delta_{jk} \\
&= A_{kj}x_j + A_{ik}x_i
\end{aligned}$$

Since i and j are dummy indexes, we can rename, for example, j to i. Then the expression simplifies to

$$\begin{aligned}
\frac{\partial}{\partial x_k}(A_{ij}x_ix_j) &= A_{ki}x_i + A_{ik}x_i \\
&= (A_{ki} + A_{ik})x_i
\end{aligned}$$

Exercises 1.2:

1. Show that $\delta_{ij}\delta_{ji} = 3$
2. Find the expression in index notation for $\frac{\partial \boldsymbol{v}}{\partial \boldsymbol{v}}$ where $\boldsymbol{v}$ is a vector.
3. With the aid of the Kronecker delta, find the expression in index notation for the trace of $\boldsymbol{a}\boldsymbol{b}^T$ where $\boldsymbol{a}$ and $\boldsymbol{b}$ are vectors
4. Simplify the expression $\frac{\partial^2}{\partial x_j \partial x_i}(x_ix_j)$

1.5 Levi-Civita symbol

The Levi-Civita symbol, or permutation operator, is a fully antisymmetric tensor, meaning that it changes sign under any permutation of its indices. In 3D space the Levi-Civita symbol is defined as

$$\epsilon_{ijk} = \begin{cases} 1 & \text{if } (i,j,k) \text{ is an even permutation of } (1,2,3) \\ -1 & \text{if } (i,j,k) \text{ is an odd permutation of } (1,2,3) \\ 0 & \text{if any two indices are the same (non-permutation)} \end{cases} \tag{1.26}$$

All the components of the Levi-Civita symbol are given by

$$\boldsymbol{\epsilon}_1 = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & -1 & 0 \end{pmatrix} \quad \boldsymbol{\epsilon}_2 = \begin{pmatrix} 0 & 0 & -1 \\ 0 & 0 & 0 \\ 1 & 0 & 0 \end{pmatrix} \quad \boldsymbol{\epsilon}_3 = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} \tag{1.27}$$

The Levi-Civita symbol has the following properties:

Symmetry: It is fully antisymmetric tensor. If any two indices are swapped, the sign of the symbol is reversed:

$$\epsilon_{ijk} = -\epsilon_{jik} \tag{1.28}$$

Even permutations: The Levi-Civita symbol is equal to 1 for even permutations of its indices

$$\epsilon_{123} = \epsilon_{231} = \epsilon_{312} = 1 \tag{1.29}$$

Odd permutations: The Levi-Civita symbol is equal to -1 for odd permutations of its indices

$$\epsilon_{132} = \epsilon_{213} = \epsilon_{321} = -1 \tag{1.30}$$

Zero for repeated indices: It is equal to zero if any two indices are the same

$$\epsilon_{112} = \epsilon_{121} = \epsilon_{122} = \epsilon_{233} = \ldots = 0 \tag{1.31}$$

Some applications of the Levi-Civita symbol include the calculation of:

Determinat: The determinant of a 3×3 matrix $\boldsymbol{A}$ is written as

$$\det(\boldsymbol{A}) = \epsilon_{ijk} A_{1i} A_{2j} A_{3k} = \epsilon_{ijk} A_{i1} A_{j2} A_{k3} \tag{1.32}$$

Cross product: The cross product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ can be written as

$$\boldsymbol{a} \times \boldsymbol{b} = \epsilon_{ijk} a_j b_k \hat{\boldsymbol{e}}_i \tag{1.33}$$

where $\hat{\boldsymbol{e}}_i$ are the basis vectors of the coordinate system.

Curl of a Vector Field: In vector calculus, the curl of a vector field $\boldsymbol{f}$ is written as

$$\nabla \times \boldsymbol{f} = \epsilon_{ijk} f_{k,j} \hat{\boldsymbol{e}}_i \tag{1.34}$$

The following identities involving the Levi-Civita symbol are useful in tensor algebra

$$\begin{aligned} \epsilon_{ijk}\epsilon_{ijk} &= 6 \\ \epsilon_{ijk}\epsilon_{mnk} &= \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm} \\ \epsilon_{imn}\epsilon_{jmn} &= 2\delta_{ij} \\ \boldsymbol{a} \cdot (\boldsymbol{b} \times \boldsymbol{c}) &= \epsilon_{ijk} a_i b_j c_k \quad \text{triple scalar product} \end{aligned} \tag{1.35}$$

Example 1.9: Show that $\epsilon_{ijk}\epsilon_{ijk} = 6$.

Solution:

Expanding for i we have

$$\epsilon_{ijk}\epsilon_{ijk} = \epsilon_{1jk}\epsilon_{1jk} + \epsilon_{2jk}\epsilon_{2jk} + \epsilon_{3jk}\epsilon_{3jk}$$

now, expanding for j and dropping terms with repeated indices

$$\begin{aligned} \epsilon_{ijk}\epsilon_{ijk} = {} & (\epsilon_{12k}\epsilon_{12k} + \epsilon_{13k}\epsilon_{13k}) + \\ & (\epsilon_{21k}\epsilon_{21k} + \epsilon_{23k}\epsilon_{23k}) + \\ & (\epsilon_{31k}\epsilon_{31k} + \epsilon_{32k}\epsilon_{32k}) \end{aligned}$$

Finally, expanding for k and simplifying we have

$$\begin{aligned} \epsilon_{ijk}\epsilon_{ijk} &= (\epsilon_{123}\epsilon_{123} + \epsilon_{132}\epsilon_{132}) + \\ & \quad (\epsilon_{213}\epsilon_{213} + \epsilon_{231}\epsilon_{231}) + \\ & \quad (\epsilon_{312}\epsilon_{312} + \epsilon_{321}\epsilon_{321}) \\ &= (1)(1) + (-1)(-1) + \\ & \quad (-1)(-1) + (1)(1) + \end{aligned}$$

$$(1)(1) + (-1)(-1)$$
$$= 6$$

Exercises 1.3: Solve the exercises below using index notation in 3D space.

1. Show that $\epsilon_{ijk} a_i a_j = 0$
2. Prove the identity $\epsilon_{ijk}\epsilon_{mnk} = \delta_{im}\delta_{jn} - \delta_{in}\delta_{jm}$
3. Using the permutation operator, prove the vector triple product identity

$$\boldsymbol{a} \times (\boldsymbol{b} \times \boldsymbol{c}) = \boldsymbol{b}(\boldsymbol{a} \cdot \boldsymbol{c}) - \boldsymbol{c}(\boldsymbol{a} \cdot \boldsymbol{b})$$

1.6 Tensor operations

Tensors can be manipulated using a variety of operations, including addition, multiplication, derivation, etc.

Addition and subtraction: Tensors of the same order can be added or subtracted by adding or subtracting their components. For example, given two second-order tensors $\boldsymbol{A}$ and $\boldsymbol{B}$, the components of $\boldsymbol{C} = \boldsymbol{A} + \boldsymbol{B}$ are given by

$$C_{ij} = A_{ij} + B_{ij} \tag{1.36}$$

Scalar multiplication: A tensor can be multiplied by a scalar, which scales all of its components. For example, given a second-order tensor $\boldsymbol{A}$ and a scalar α, the components of the product $\boldsymbol{C} = \alpha\boldsymbol{A}$ are given by

$$C_{ij} = \alpha A_{ij} \tag{1.37}$$

Inner product: The inner product (or contraction) between two tensors sums over one or more matching indices, reducing the rank of the resulting tensor. For example, the inner product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is given by

$$\boldsymbol{a} \cdot \boldsymbol{b} = a_i b_i \tag{1.38}$$

This reduces the tensors' rank, producing a scalar, and is analogous to the dot product for vectors.

For the inner product of a second-order tensor $\boldsymbol{A}$ and a vector $\boldsymbol{b}$, the result is a vector given by

$$\boldsymbol{c} = \boldsymbol{A} \cdot \boldsymbol{b} = \boldsymbol{A}\boldsymbol{b} \tag{1.39}$$

where the components of the resulting vector $\boldsymbol{c}$ are given by

$$c_i = A_{ij} b_j \tag{1.40}$$

For the case of two second-order tensors $\boldsymbol{A}$ and $\boldsymbol{B}$, the inner product is a scalar quantity that is given by

$$\boldsymbol{A} : \boldsymbol{B} = A_{ij} B_{ij} \tag{1.41}$$

The operator $:$ is called double contraction and is equivalent to the Frobenius inner product.

Similarly as the dot product between two vectors is related to the angle between those vectors, the double contraction product between two second-order tensors is related to the degree of alignment between them.

Outter product: The outer product of two tensors results in a tensor of higher rank. For example, the outer product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ results in a second-order tensor $\boldsymbol{C} = \boldsymbol{a} \otimes \boldsymbol{b}$, where the components are given by

$$C_{ij} = a_i b_j \tag{1.42}$$

For the case of two second-order tensors $\boldsymbol{A}$ and $\boldsymbol{B}$, the outer product results in a fourth-order tensor $\boldsymbol{C} = \boldsymbol{A} \otimes \boldsymbol{B}$ with components

$$C_{ijkl} = A_{ij} B_{kl} \tag{1.43}$$

Dyadic product: The dyadic product is a specific case of the outer product applied to vectors to form a second-order tensor. For example, the dyadic product of two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is given by

$$\boldsymbol{C} = \boldsymbol{a} \otimes \boldsymbol{b} = \boldsymbol{a}\boldsymbol{b}^T \tag{1.44}$$

Cross product: In three-dimensional space, the cross product of two vectors results in another vector that is perpendicular to both input vectors. Using the Levi-Civita symbol, the cross product of vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ is

$$\boldsymbol{a} \times \boldsymbol{b} = \epsilon_{ijk} a_j b_k \hat{\boldsymbol{e}}_i \tag{1.45}$$

Tensor contraction: Tensor contraction is an operation that reduces the rank of a tensor by summing over a pair of matching indices. For instanace, contracting a second-order tensor $\boldsymbol{A}$ with components A_{ij} over its indices yields a scalar

$$\begin{aligned} A_{ii} &= \sum_{i=1}^{3} A_{ii} \\ &= A_{ij}\delta_{ij} \\ &= \boldsymbol{A}:\boldsymbol{I} \end{aligned} \tag{1.46}$$

This operation is the same as taking the trace of a matrix. Note that the Kronecker delta can be used to perform the contraction of the expression A_{ij}.

Another example of tensor contraction is the operation called tensor acting on a vector. Given a second-order tensor $\boldsymbol{A}$ and a vector $\boldsymbol{b}$, the result is a vector $\boldsymbol{c} = \boldsymbol{A} \cdot \boldsymbol{b} = \boldsymbol{A}\boldsymbol{b}$ with components

$$c_i = A_{ij} b_j \tag{1.47}$$

This operation is a generalization of matrix-vector multiplication and is used extensively to model the transformation of vectors under linear operations

Derivative: The derivative of a tensor with respect to a variable is a tensor whose components are the derivatives of the original tensor's components. For example, the derivative of a scalar function f with respect to $\boldsymbol{x} = (x_1, x_2, x_3)$ is the gradient of f given by

$$\nabla f = \begin{pmatrix} \frac{\partial f}{\partial x_1} \\ \frac{\partial f}{\partial x_2} \\ \frac{\partial f}{\partial x_3} \end{pmatrix} \tag{1.48}$$

where each component is given by

$$f_{,i} = \frac{\partial f}{\partial x_i} \tag{1.49}$$

Similarly, the derivative of a vector field $\boldsymbol{f}$ with respect to $\boldsymbol{x}$, gradient of a vector field, is a second-order tensor field given by

$$f_{i,j} = \frac{\partial f_i}{\partial x_j} \tag{1.50}$$

In turn, the divergence of a vector field $\boldsymbol{f}$ results in a scalar field and is given by

$$\begin{aligned} \nabla \cdot \boldsymbol{f} &= \frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} + \frac{\partial f_3}{\partial x_3} \\ &= f_{i,i} \end{aligned} \tag{1.51}$$

The derivative of a second-order tensor $\boldsymbol{A}$ with respect to itselft provides a fourth-order tensor whith components

$$\frac{\partial A_{ij}}{\partial A_{mn}} = \delta_{im}\delta_{jn} \tag{1.52}$$

This means the derivative is zero unless both indices match, in which case it is 1. The Kronecker delta ensures that the result is 1 only when the indices align.

Example 1.10: Shown that for a symmetric second-order tensor $\boldsymbol{A}$, the following holds: $\text{trace}(\boldsymbol{A}^2) = \boldsymbol{A}:\boldsymbol{A}$.

Solution:

Using index notation the components of $\boldsymbol{A}^2$ are

$$\left(\boldsymbol{A}^2\right)_{ik} = A_{ij}A_{jk}$$

To find the trace, we perform a contraction voert the indices i and k

$$\begin{aligned} \text{trace}(\boldsymbol{A}^2) &= A_{ij}A_{jk}\delta_{ki} \\ &= A_{ij}A_{ji} \end{aligned}$$

Since $\boldsymbol{A}$ is symmetric, we have $\text{trace}(\boldsymbol{A}^2) = A_{ij}A_{ij}$.

On the other hand, the double contraction $\boldsymbol{A}:\boldsymbol{A}$ is given by

$$\boldsymbol{A}:\boldsymbol{A} = A_{ij}A_{ij}$$

Therefore, we conclude that $\text{trace}(\boldsymbol{A}^2) = \boldsymbol{A}:\boldsymbol{A}$.

Example 1.11: Given a second-order tensor $\boldsymbol{A}$ calculate $\frac{\partial}{\partial \boldsymbol{A}}(\text{trace}(\boldsymbol{A}))$.

Solution:

Since the trace of $\boldsymbol{A}$ is a scalar, the result of the derivative is a second-order tensor. Using the summation convention, the trace of $\boldsymbol{A}$ is given by A_{kk}. Then, its derivative with respect to the components of $\boldsymbol{A}$ can be expressed conveniently in terms of the Kronecker delta as

$$\frac{\partial A_{kk}}{\partial A_{ij}} = \frac{\partial A_{kn}\delta_{nk}}{\partial A_{ij}}$$

Performing the derivative

$$\begin{aligned}\frac{\partial A_{kk}}{\partial A_{ij}} &= \delta_{ki}\delta_{nj}\delta_{nk} \\ &= \delta_{ni}\delta_{nj} \\ &= \delta_{ij}\end{aligned}$$

Then, we can conlude that the derivative of the trace of a second-order tensor $\boldsymbol{A}$ with respect to $\boldsymbol{A}$ is the second-order identity tensor

$$\frac{\partial\, \text{trace}(\boldsymbol{A})}{\partial \boldsymbol{A}} = \boldsymbol{I}$$

Example 1.12: Given two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, calculate $\frac{\partial}{\partial \boldsymbol{a}}(\boldsymbol{a} \cdot \boldsymbol{b})$.

The components of the inner product $\boldsymbol{a} \cdot \boldsymbol{b}$ ca be expressed as $a_j b_j$. Then, the derivative of this expression with respect to the components of $\boldsymbol{a}$ is

$$\begin{aligned}\frac{\partial}{\partial a_i}(a_j b_j) &= \frac{\partial a_j}{\partial a_i} b_j \\ &= \delta_{ji} b_j \\ &= b_i\end{aligned}$$

Then, we can conclude that $\frac{\partial}{\partial \boldsymbol{a}}(\boldsymbol{a} \cdot \boldsymbol{b}) = \boldsymbol{b}$

Example 1.13: Given a vectors $\boldsymbol{v}$, calculate $\frac{\partial \|\boldsymbol{a}\|}{\partial \boldsymbol{a}}$.

Solution:

The norm of $\boldsymbol{a}$ can be written as $\|\boldsymbol{a}\| = \sqrt{\boldsymbol{a} \cdot \boldsymbol{a}}$. The componets of the norm are given by $\|\boldsymbol{a}\| = \sqrt{a_j a_j}$. Then, the components $\frac{\partial \|\boldsymbol{a}\|}{\partial \boldsymbol{a}}$ can be expressed in index notation as

$$\begin{aligned}\frac{\partial \|\boldsymbol{a}\|}{\partial \boldsymbol{a}} &= \frac{\partial \sqrt{a_j a_j}}{\partial \boldsymbol{a}} \\ &= \frac{\partial \sqrt{a_j a_j}}{\partial a_i} \\ &= \frac{1}{2}(a_j a_j)^{-\frac{1}{2}} \frac{\partial a_k a_k}{\partial a_i}\end{aligned}$$

Note that the dummy index k was introduced to avoid two pairs of the index j. Then, applying the product rule we have

$$\begin{aligned}\frac{\partial \|\boldsymbol{a}\|}{\partial \boldsymbol{a}} &= \frac{1}{2}(a_j a_j)^{-\frac{1}{2}} \left(\frac{\partial a_k}{\partial a_i} a_k + a_k \frac{\partial a_k}{\partial a_i}\right) \\ &= \frac{1}{2}(a_j a_j)^{-\frac{1}{2}} (\delta_{ki} a_k + a_k \delta_{ki})\end{aligned}$$

$$= \frac{1}{2}(a_j a_j)^{-\frac{1}{2}}(a_i + a_i)$$
$$= \frac{a_i}{\sqrt{a_j a_j}}$$

Then, in abstract notation, we can write that the derivative of the norm of a vector with respect to itself is equal to the unit vector

$$\frac{\partial \|\boldsymbol{a}\|}{\partial \boldsymbol{a}} = \frac{\boldsymbol{a}}{\|\boldsymbol{a}\|} = \hat{\boldsymbol{a}}$$

Exercises 1.4:

1. Given two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$, show that $(\boldsymbol{a} \otimes \boldsymbol{a})(\boldsymbol{b} \otimes \boldsymbol{b}) = (\boldsymbol{a} \cdot \boldsymbol{b})\boldsymbol{a} \otimes \boldsymbol{b}$
2. Find the abstract expression for $\frac{\partial}{\partial \boldsymbol{v}}(\boldsymbol{v} \cdot \boldsymbol{v})$
3. Given two second-order tensor $\boldsymbol{A}$ and $\boldsymbol{B}$, show that $\frac{\partial}{\partial \boldsymbol{A}}(\boldsymbol{A} : \boldsymbol{B}) = \boldsymbol{B}$.
4. Find the expression for $\frac{\partial}{\partial \boldsymbol{a}}(\boldsymbol{a} \times \boldsymbol{b})$
5. Compute the derivative of $\boldsymbol{A} - \frac{1}{3}\text{trace}(\boldsymbol{A})\boldsymbol{I}$ with respect to $\boldsymbol{A}$.

1.7 Identity tensors

An identity tensor is a mathematical entity that preserves the structure of tensors during operations, acting as the tensorial equivalent of an identity matrix. It maps tensors of a specific order onto themselves without altering their components.

1.7.1 Second-order identity tensor

The second-order identity tensor, often denoted as $\boldsymbol{I}$, satisfies:

$$\boldsymbol{I} \cdot \boldsymbol{A} = \boldsymbol{A} \cdot \boldsymbol{I} = \boldsymbol{A} \tag{1.53}$$

for any second-order tensor $\boldsymbol{A}$, and

$$\boldsymbol{I} \cdot \boldsymbol{v} = \boldsymbol{v} \tag{1.54}$$

for any vector $\boldsymbol{v}$.

In index notation, it is expressed as

$$I_{ij} = \delta_{ij} \tag{1.55}$$

1.7.2 Fourth-order identity tensor

Analogous to the second-order identity tensor, the fourth-order identity tensor $\mathbb{I}$ is used in operations involving second-order tensors. It satisfies

$$\mathbb{I} : \boldsymbol{A} = \boldsymbol{A} : \mathbb{I} = \boldsymbol{A} \tag{1.56}$$

for any second-order tensor $\boldsymbol{A}$.

In index notation, it is expressed as:

$$\mathbb{I}_{ijkl} = \delta_{ik}\delta_{jl} \tag{1.57}$$

1.8 Symmetric and antisymmetric projection tensors

The fourth-order identity tensor can be decomposed into symmetric and antisymmetric parts

$$\mathbb{I} = \mathbb{I}^{\text{sym}} + \mathbb{I}^{\text{anti}} \tag{1.58}$$

These tensors, ($\mathbb{I}^{\text{sym}}$ and $\mathbb{I}^{\text{anti}}$), are known as the symmetric projection tensor and the antisymmetric projection tensor, respectively. They are used to decompose a second-order tensor into its symmetric and antisymmetric components. The components of these projection tensors are given by

$$\begin{aligned} \mathbb{I}^{\text{sym}}_{ijkl} &= \frac{1}{2}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right) \\ \mathbb{I}^{\text{anti}}_{ijkl} &= \frac{1}{2}\left(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}\right) \end{aligned} \tag{1.59}$$

For any second-order tensor $\boldsymbol{A}$, the decomposition is expressed as

$$\boldsymbol{A} = \boldsymbol{A}^{\text{sym}} + \boldsymbol{A}^{\text{anti}} \tag{1.60}$$

where

$$\begin{aligned} \boldsymbol{A}^{\text{sym}} &= \mathbb{I}^{\text{sym}} : \boldsymbol{A} \\ \boldsymbol{A}^{\text{anti}} &= \mathbb{I}^{\text{anti}} : \boldsymbol{A} \end{aligned} \tag{1.61}$$

Example 1.14: Find the symmetric and antisymmetric parts of the tensor

$$\boldsymbol{A} = \begin{pmatrix} 1 & 2 & 3 \\ 0 & 1 & 4 \\ 5 & 6 & 0 \end{pmatrix}$$

Solution: Using the fourth-order symmetric projection tensor, the symmetric part of $\boldsymbol{A}$ is given by

$$\boldsymbol{A}^{\text{sym}} = \mathbb{I}^{\text{sym}} : \boldsymbol{A}$$

where the components of $\boldsymbol{A}^{\text{sym}}$ are given by

$$A^{\text{sym}}_{ij} = \frac{1}{2}\left(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}\right)A_{kl}$$

Performing the contraction we have

$$A^{\text{sym}}_{ij} = \frac{1}{2}\left(A_{ij} + A_{ji}\right)$$

Then, the components of the symmetric part of $\boldsymbol{A}$ are

$$\boldsymbol{A}^{\text{sym}} = \begin{pmatrix} 1 & 1 & 4 \\ 1 & 1 & 5 \\ 4 & 5 & 0 \end{pmatrix}$$

Similarly, using the fourth-order antisymmetric projection tensor, the antisymmetric part of $\boldsymbol{A}$ is given by

$$\boldsymbol{A}^{\text{anti}} = \mathbb{I}^{\text{anti}} : \boldsymbol{A}$$

where the components of $\boldsymbol{A}^{\text{anti}}$ are given by

$$A_{ij}^{\text{anti}} = \frac{1}{2}\big(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}\big)A_{kl}$$

Performing the contraction we have

$$A_{ij}^{\text{anti}} = \frac{1}{2}\big(A_{ij} - A_{ji}\big)$$

Then, the components of the antisymmetric part of $\boldsymbol{A}$ are

$$\boldsymbol{A}^{\text{anti}} = \begin{pmatrix} 0 & 1 & -1 \\ -1 & 0 & -1 \\ 1 & 1 & 0 \end{pmatrix}$$

Exercises 1.5:

1. Using index notation, show that
   a) $\boldsymbol{I}:\boldsymbol{I} = 3$
   b) $\mathbb{I}:\mathbb{I} = 3\boldsymbol{I}$
   c) $\mathbb{I}^{\text{sym}}:\mathbb{I}^{\text{sym}} = \mathbb{I}^{\text{sym}}$
2. Considering the second-order tensor $\boldsymbol{A}$, using index notation, show that
   a) $(\boldsymbol{I}\otimes\boldsymbol{I}):\boldsymbol{A} = \text{trace}(\boldsymbol{A})\,\boldsymbol{I}$
   b) $\boldsymbol{A}^{\text{anti}} = \frac{1}{2}(\boldsymbol{A} - \boldsymbol{A}^T)$

1.9 Cartesian basis vectors

In the context of Cartesian (rectangular) coordinates, the unit vectors are the standard basis vectors that point along the coordinate axes. In 3D space, the Cartesian basis vectors are denoted as $\hat{\boldsymbol{e}}_1$, $\hat{\boldsymbol{e}}_2$, and $\hat{\boldsymbol{e}}_3$. In a right-handed coordinate system, $\hat{\boldsymbol{e}}_{\mathbf{1}}$ is the unit vector along the x-axis, $\hat{\boldsymbol{e}}_{\mathbf{2}}$ is the unit vector along the y-axis, and $\hat{\boldsymbol{e}}_{\mathbf{3}}$ is the unit vector along the z-axis and are given by

$$\hat{\boldsymbol{e}}_{\mathbf{1}} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix}, \quad \hat{\boldsymbol{e}}_{\mathbf{2}} = \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}, \quad \hat{\boldsymbol{e}}_{\mathbf{3}} = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \tag{1.62}$$

An arbitrary vector $\boldsymbol{v}$ can be expressed as a linear combination of the basis vectors as

$$\boldsymbol{v} = v_1\hat{\boldsymbol{e}}_{\mathbf{1}} + v_2\hat{\boldsymbol{e}}_{\mathbf{2}} + v_3\hat{\boldsymbol{e}}_{\mathbf{3}} \tag{1.63}$$

where v_1, v_2, and v_3 are the components of the vector $\boldsymbol{v}$ along the x_1, x_2, and x_3 axes, respectively. This representation is unique in a Cartesian coordinate system due to the linear independence of the basis vectors.

For a second-order tensor $\boldsymbol{A}$, it can be expressed as

$$\begin{aligned} \boldsymbol{A} &= \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \\ &= A_{11}\begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + A_{12}\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + A_{13}\begin{pmatrix} 0 & 0 & 1 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \cdots + A_{33}\begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \end{aligned} \tag{1.64}$$

which, can be condensed as

$$\boldsymbol{A} = A_{ij}\hat{\boldsymbol{e}}_{\boldsymbol{i}} \otimes \hat{\boldsymbol{e}}_{\boldsymbol{j}} \tag{1.65}$$

Therefore, the second-order tensors $\hat{\boldsymbol{e}}_{\boldsymbol{i}} \otimes \hat{\boldsymbol{e}}_{\boldsymbol{j}}$ represent the basis tensors in a Cartesian coordinate system.

The cartesian basis vectors have the following properties:

Ortogonality: The basis vectors are orthogonal to each other, meaning that their dot products are zero:

$$\hat{\boldsymbol{e}}_{\boldsymbol{1}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{2}} = 0, \quad \hat{\boldsymbol{e}}_{\boldsymbol{2}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{3}} = 0, \quad \hat{\boldsymbol{e}}_{\boldsymbol{1}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{3}} = 0 \tag{1.66}$$

Orthonormality: Since they are both orthogonal and normalized, the Cartesian basis vectors form an orthonormal basis. This means they can represent any vector in space without any ambiguity or overlap between the coordinate directions.

The cross product of two basis vectors is given by the Levi-Civita symbol

$$\hat{\boldsymbol{e}}_i \times \hat{\boldsymbol{e}}_j = \varepsilon_{ijk}\hat{\boldsymbol{e}}_k \tag{1.67}$$

For example

$$\begin{aligned} \hat{\boldsymbol{e}}_1 \times \hat{\boldsymbol{e}}_2 &= \hat{\boldsymbol{e}}_3 \\ \hat{\boldsymbol{e}}_2 \times \hat{\boldsymbol{e}}_3 &= \hat{\boldsymbol{e}}_1 \\ \hat{\boldsymbol{e}}_2 \times \hat{\boldsymbol{e}}_1 &= -\hat{\boldsymbol{e}}_3 \end{aligned} \tag{1.68}$$

Also, the following identity is valid

$$\hat{\boldsymbol{e}}_{\boldsymbol{i}} \cdot \hat{\boldsymbol{e}}_{\boldsymbol{j}} = \delta_{ij} \tag{1.69}$$

Some tensor operations can be expressed in terms of the basis vectors. For example, the dot product between two vectors $\boldsymbol{a}$ and $\boldsymbol{b}$ can be written as

$$\boldsymbol{a} \cdot \boldsymbol{b} = (a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2 + a_3\hat{\boldsymbol{e}}_3) \cdot (b_1\hat{\boldsymbol{e}}_1 + b_2\hat{\boldsymbol{e}}_2 + b_3\hat{\boldsymbol{e}}_3) = a_1b_1 + a_2b_2 + a_3b_3 \tag{1.70}$$

Similarly, the cross product between those vectors is given by

$$\begin{aligned} \boldsymbol{a} \times \boldsymbol{b} &= (a_1\hat{\boldsymbol{e}}_1 + a_2\hat{\boldsymbol{e}}_2 + a_3\hat{\boldsymbol{e}}_3) \times (b_1\hat{\boldsymbol{e}}_1 + b_2\hat{\boldsymbol{e}}_2 + b_3\hat{\boldsymbol{e}}_3) \\ &= (a_2b_3 - a_3b_2)\hat{\boldsymbol{e}}_1 + (a_3b_1 - a_1b_3)\hat{\boldsymbol{e}}_2 + (a_1b_2 - a_2b_1)\hat{\boldsymbol{e}}_3 \end{aligned} \tag{1.71}$$

or

$$\boldsymbol{a} \times \boldsymbol{b} = \epsilon_{ijk}a_jb_k\hat{\boldsymbol{e}}_i \tag{1.72}$$

The gradient of a scalar function f can be expressed in terms of the basis vectors as

$$\begin{aligned} \nabla f &= \frac{\partial f}{\partial x_1}\hat{\boldsymbol{e}}_1 + \frac{\partial f}{\partial x_2}\hat{\boldsymbol{e}}_2 + \frac{\partial f}{\partial x_3}\hat{\boldsymbol{e}}_3 \\ &= f_{,i}\hat{\boldsymbol{e}}_i \end{aligned} \tag{1.73}$$

In the same way, the curl of a vector field $\boldsymbol{f}$ is given by

$$\begin{aligned} \nabla \times \boldsymbol{f} &= \left(f_{3,2} - f_{2,3}\right)\hat{\boldsymbol{e}}_1 + \left(f_{1,3} - f_{3,1}\right)\hat{\boldsymbol{e}}_2 + \left(f_{2,1} - f_{1,2}\right)\hat{\boldsymbol{e}}_3 \\ &= \epsilon_{ijk}f_{k,j}\hat{\boldsymbol{e}}_i \end{aligned} \tag{1.74}$$

1.10 Coordinates transformation

Coordinate transformations are fundamental tools used in physics, mathematics, and engineering to change the representation of vectors, tensors, and other geometric objects from one coordinate system to another. These transformations allow us to describe the same physical phenomena or mathematical objects from different perspectives or in different frames of reference.

A coordinate transformation is a rule or a function that relates the coordinates of a point in one coordinate system to the coordinates of the same point in another coordinate system.

Given two coordinate systems (say, old and new), we may call the old coordinate system $\boldsymbol{x} = (x_1, x_2, x_3)$ and the new coordinate system $\boldsymbol{x}' = (x_1', x_2', x_3')$. The transformation provides a mathematical relationship between (x_1, x_2, x_3) and (x_1', x_2', x_3').

There are different types of coordinate transformations

Linear transformations: A linear transformation is a mathematical operation that transforms a vector (or tensor) into another vector (or tensor). For example, a rotation, scaling, or reflection of the coordinate system. The new coordinates are linear combinations of the old coordinates.

A transformation $\boldsymbol{T}$ is called linear if it satisfies the following properties

$$\begin{aligned} \boldsymbol{T}(\boldsymbol{a}+\boldsymbol{b}) &= \boldsymbol{T}\boldsymbol{a} + \boldsymbol{T}\boldsymbol{b} \\ \boldsymbol{T}(\alpha\boldsymbol{a}) &= \alpha\boldsymbol{T}\boldsymbol{a} \end{aligned} \tag{1.75}$$

Non-linear transformations: These are transformations that cannot be represented by a matrix multiplication, it means that the new coordinates are not linear combinations of the old coordinates. For example, a polar to Cartesian coordinate transformation.

Rigid transformations: These are transformations that preserve the distance between points. They include translations, rotations and reflections. These transformations preserve the shape and size of objects in the space.

Affine transformations: Affine transformations are more general than rigid transformations. These are transformations that preserve the straight lines and planes. They include translations, rotations, reflections, scaling and shearing.

Curviliear transformations: These are transformations that map straight lines in the old coordinate system to curves in the new coordinate system. For example, a transformation from Cartesian to spherical coordinates.

1.11 Tensor transformation law

This is a linear transformation that allows to express the components of a tensor in a new coordinate system (x_1', x_2', x_3') based on its components in the original system (x_1, x_2, x_3) and the relationship between the two coordinate systems. For this purpose, we define a second-order transformation tensor $\boldsymbol{R}$ so the components of the unit vectors that define the new coordinate system can be expressed as

$$\hat{\boldsymbol{e}}_i' = \boldsymbol{R}\hat{\boldsymbol{e}}_i \tag{1.76}$$

This expression can be extended into a product of second-order tensors, so all components from the unit vectors are taken into account

$$\begin{pmatrix} | & | & | \\ \hat{\boldsymbol{e}}'_1 & \hat{\boldsymbol{e}}'_2 & \hat{\boldsymbol{e}}'_3 \\ | & | & | \end{pmatrix} = \begin{pmatrix} R_{11} & R_{12} & R_{13} \\ R_{21} & R_{22} & R_{23} \\ R_{31} & R_{32} & R_{33} \end{pmatrix} \underbrace{\begin{pmatrix} | & | & | \\ \hat{\boldsymbol{e}}_1 & \hat{\boldsymbol{e}}_2 & \hat{\boldsymbol{e}}_3 \\ | & | & | \end{pmatrix}}_{\boldsymbol{I}} \tag{1.77}$$

Since the tensor composed by the basis vectors of the original coordinate system is equal to the identity tensor, we can conclude that the componentes from the transformation tensor $\boldsymbol{R}$ are the components of the new basis vectors in the old coordinate system. So, we may write

$$\boldsymbol{R} = \begin{pmatrix} | & | & | \\ \hat{\boldsymbol{e}}'_1 & \hat{\boldsymbol{e}}'_2 & \hat{\boldsymbol{e}}'_3 \\ | & | & | \end{pmatrix} \tag{1.78}$$

The components of $\boldsymbol{R}$ can be found considering that the directions of $\hat{\boldsymbol{e}}'_1$, $\hat{\boldsymbol{e}}'_2$, $\hat{\boldsymbol{e}}'_3$ are given by the direction cosines, $\cos(x_i, x'_j)$, between the basis vectors of the new and old coordinate systems. Thus

$$\boldsymbol{R} = \begin{pmatrix} \cos(x_1, x'_1) & \cos(x_1, x'_2) & \cos(x_1, x'_3) \\ \cos(x_2, x'_1) & \cos(x_2, x'_2) & \cos(x_2, x'_3) \\ \cos(x_3, x'_1) & \cos(x_3, x'_2) & \cos(x_3, x'_3) \end{pmatrix} \tag{1.79}$$

This is the same as expressing the components using the dot product between the basis vectors as

$$R_{ij} = \hat{\boldsymbol{e}}_i \cdot \hat{\boldsymbol{e}}'_j \tag{1.80}$$

Note that $\boldsymbol{R}$ is orthogonal, meaning that its inverse is equal to its transpose. Also, note that the basis vectors of the new coordinate system expressed in the old coordinate system can be written as

$$\begin{aligned} \hat{\boldsymbol{e}}'_1 &= R_{11}\hat{\boldsymbol{e}}_1 + R_{21}\hat{\boldsymbol{e}}_2 + R_{31}\hat{\boldsymbol{e}}_3 \\ \hat{\boldsymbol{e}}'_2 &= R_{12}\hat{\boldsymbol{e}}_1 + R_{22}\hat{\boldsymbol{e}}_2 + R_{32}\hat{\boldsymbol{e}}_3 \\ \hat{\boldsymbol{e}}'_3 &= R_{13}\hat{\boldsymbol{e}}_1 + R_{23}\hat{\boldsymbol{e}}_2 + R_{33}\hat{\boldsymbol{e}}_3 \end{aligned} \tag{1.81}$$

These equations can be synthesized as

$$\hat{\boldsymbol{e}}'_i = R_{ji}\hat{\boldsymbol{e}}_j \tag{1.82}$$

For an arbitrary vector $\boldsymbol{v}$, the components in the new coordinate system are given by

$$\begin{aligned} v'_i &= \hat{\boldsymbol{e}}'_i \cdot \boldsymbol{v} \\ &= R_{ji}\hat{\boldsymbol{e}}_j \cdot \boldsymbol{v} \\ &= R_{ji}v_j \end{aligned} \tag{1.83}$$

that represents the following operation in abstract notation

$$\boldsymbol{v}' = \boldsymbol{R}^T\boldsymbol{v} \tag{1.84}$$

For a second-order tensor $\boldsymbol{A}$, its components in the new coordinate system are given by

$$
\begin{aligned}
A'_{ij} &= \hat{\boldsymbol{e}}'_i \cdot \boldsymbol{A}\hat{\boldsymbol{e}}'_j \\
&= R_{mi}\hat{\boldsymbol{e}}_m \cdot \boldsymbol{A}R_{ni}\hat{\boldsymbol{e}}_n \\
&= R_{mi}R_{ni}\hat{\boldsymbol{e}}_m \cdot \boldsymbol{A}\hat{\boldsymbol{e}}_n \\
&= R_{mi}R_{ni}A_{mn}
\end{aligned}
\tag{1.85}
$$

Then, the tensor in the new coordinate system can be expressed in abstract notation as

$$
\boldsymbol{A}' = \boldsymbol{R}^T\boldsymbol{A}\boldsymbol{R} \tag{1.86}
$$

For different orders, the transformation rules are different. For instance

$$
\begin{aligned}
\alpha' &= \alpha \\
a'_i &= R_{mi}a_m \\
A'_{ij} &= R_{mi}R_{nj}A_{mn} \\
A'_{ijk} &= R_{mi}R_{nj}R_{pk}A_{mnp} \\
A'_{ijkl} &= R_{mi}R_{nj}R_{pk}R_{ql}A_{mnpq}
\end{aligned}
\tag{1.87}
$$

Example 1.15: Given a vector $\boldsymbol{v}$ and a second-order tensor $\boldsymbol{A}$, find the corresponding components in a new coordinate system which is rotated 53.13° around the z-axis.

$$
\boldsymbol{v} = \begin{pmatrix} 2 \\ 3 \\ 4 \end{pmatrix} \qquad \boldsymbol{A} = \begin{pmatrix} 1 & 2 & 0 \\ 2 & 4 & 1 \\ 0 & 1 & 3 \end{pmatrix}
$$

Solution:

For a rotation of an angle $\theta = 53.13°$ around the z-axis, the transformation matrix takes the form

$$
\boldsymbol{R} = \begin{pmatrix} \cos(\theta) & -\sin(\theta) & 0 \\ \sin(\theta) & \cos(\theta) & 0 \\ 0 & 0 & 1 \end{pmatrix} = \begin{pmatrix} 0.6 & -0.8 & 0 \\ 0.8 & 0.6 & 0 \\ 0 & 0 & 1 \end{pmatrix}
$$

The transformation of vector $\boldsymbol{v}$ can be performed using the rule

$$
v'_i = R_{ji}v_j
$$

or, using direct notation

$$
\boldsymbol{v}' = \boldsymbol{R}^T\boldsymbol{v}
$$

Then, the resulting vector $\boldsymbol{v}'$ is given by

$$
\boldsymbol{v}' = \begin{pmatrix} 3.6 \\ 0.2 \\ 4 \end{pmatrix}
$$

Similarly, the transformation of tensor $\boldsymbol{A}$ can be performed using the rule

$$
A'_{ij} = R_{mi}R_{nj}A_{mn}
$$

or, using direct notation

$$\boldsymbol{A}' = \boldsymbol{R}^T \boldsymbol{A} \boldsymbol{R}$$

Then, the resulting tensor $\boldsymbol{A}'$ is given by

$$\boldsymbol{A}' = \begin{pmatrix} 4.84 & 0.88 & 0.8 \\ 0.88 & 0.16 & 0.6 \\ 0.8 & 0.6 & 3 \end{pmatrix}$$

Exercises 1.6:

1. Given the second-order tensor

$$\boldsymbol{A} = \begin{pmatrix} 3 & 0 & -1 \\ 0 & 4 & 0 \\ -1 & 0 & 2 \end{pmatrix}$$

find the components of the tensor in a new coordinate system which is

a) rotated 36.87° around the y-axis.

b) rotated 45° around the z-axis.

1.12 Eigenvalues and eigenvectors

For a second-order tensor, the eigenvalues and eigenvectors of a second-order tensor are fundamental concepts that describe the intrinsic properties of the tensor. Given a second-order tensor $\boldsymbol{A}$, the eigenvalue problem consists of finding the eigenvalues λ_i and corresponding eigenvectors $\boldsymbol{v}_i$ that satisfy the following equation

$$\boldsymbol{A}\boldsymbol{v} = \lambda \boldsymbol{v} \tag{1.88}$$

This equation states that when the tensor $\boldsymbol{A}$ acts on a vector $\boldsymbol{v}$, the result is a scaled version of $\boldsymbol{v}$, where the scaling factor is λ.

Rearranging the equation above, we have

$$(\boldsymbol{A} - \lambda \boldsymbol{I})\boldsymbol{v} = 0 \tag{1.89}$$

where $\boldsymbol{I}$ is the identity tensor. For a non-trivial solution (i.e. $\boldsymbol{v} \neq 0$), the determinant of $\boldsymbol{A} - \lambda \boldsymbol{I}$ must be zero

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0 \tag{1.90}$$

This equation is called the characteristic equation, and solving it gives the eigenvalues λ_i of the tensor.

Expanding the determinant gives a cubic equation

$$\lambda^3 - I_1 \lambda^2 + I_2 \lambda - I_3 = 0 \tag{1.91}$$

where I_1, I_2 and I_3 are called principal invariants of $\boldsymbol{A}$ and the roots λ_1, λ_2 and λ_3 are the eigenvalues of the tensor. The roots can be found by applying numerical methods or by using analytical solutions as the Cardano's formula. For symmetric tensors, the following solution based on the Cardano's formula can be used

$$\lambda_i = \frac{1}{3}\left(I_1 + 2\sqrt{I_1^2 - 3I_2} \cos\left(\frac{\theta + 2\pi i}{3} \right) \right) \quad i = 1, 2, 3 \tag{1.92}$$

where θ is given by

$$\theta = \cos^{-1}\left(\frac{2I_1^3 - 9I_1I_2 + 27I_3}{2(I_1^2 - 3I_2)^{\frac{3}{2}}}\right) \tag{1.93}$$

Once the eigenvalues are found, the corresponding eigenvectors can be determined by solving the equation

$$(\boldsymbol{A} - \lambda_i \boldsymbol{I})\boldsymbol{v}_i = \boldsymbol{0} \tag{1.94}$$

The eigenvectors represent the directions in which the tensor acts as a simple stretch or compression. For symmetric tensors, they represent a set of orthogonal directions along which the tensor has no shear deformation, as shown in Figure 1.1. These directions are called the principal directions of the tensor.

The transformation required to express the tensor components in a coordinate system aligned with the principal directions is given by

$$A'_{ij} = V_{mi}V_{nj}A_{mn} \tag{1.95}$$

that can be written in abstract notation as

$$\boldsymbol{A}' = \boldsymbol{V}^T\boldsymbol{A}\boldsymbol{V} \tag{1.96}$$

where $\boldsymbol{V}$ is the second-order tensor where the columns are the eigenvectors of $\boldsymbol{A}$, thus

$$\boldsymbol{V} = \begin{pmatrix} | & | & | \\ \boldsymbol{v}_1 & \boldsymbol{v}_2 & \boldsymbol{v}_3 \\ | & | & | \end{pmatrix} \tag{1.97}$$

Example 1.16: Given a second-order tensor $\boldsymbol{A}$

$$\boldsymbol{A} = \begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & -4 \end{pmatrix}$$

calculate the eigenvalues and eigenvectors.

Solution:

The eigenvalues of the tensor can be found by solving the characteristic equation

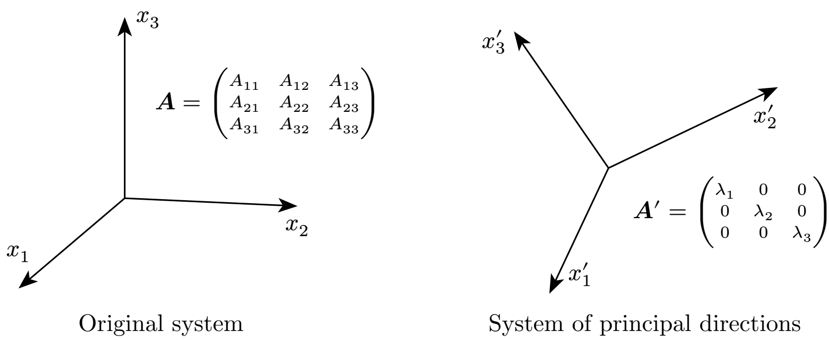

Figure 1.1: Tensor components in the system of principal directions

$$\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$$

which results in the cubic equation

$$\lambda^3 - 2\lambda^2 - 16\lambda + 32 = 0$$

The eigenvalues are given by the roots of this equation that can be found using the analytical solution as $\lambda_1 = -4$, $\lambda_2 = 2$, and $\lambda_3 = 4$. The eigenvectors can be found by solving the equation

$$(\boldsymbol{A} - \lambda_i \boldsymbol{I})\boldsymbol{v}_i = \boldsymbol{0}$$

for each eigenvalue. For example, for find the third eigenvector we have

$$\left(\begin{pmatrix} 3 & 1 & 0 \\ 1 & 3 & 0 \\ 0 & 0 & -4 \end{pmatrix} - 4 \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \right) \begin{pmatrix} v_{11} \\ v_{12} \\ v_{13} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$
$$\begin{pmatrix} -1 & 1 & 0 \\ 1 & -1 & 0 \\ 0 & 0 & -8 \end{pmatrix} \begin{pmatrix} v_{11} \\ v_{12} \\ v_{13} \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}$$

In this system, the rank of the coefficients matrix is 2 wich is less than the number of unknowns. Therefore, the system is syngular and has an infinite sets of solutions, i.e., there is an infinite number of eigenvectors associated with each eigenvalue. To find one set of solutions, one of the components of the eigenvector can be chosen arbitrarily. Let's choose $v_{11} = 1$ and solve for the remaining components. In this case, we find $v_{12} = 1$ and $v_{13} = 0$. The same procedure can be applied to find the remaining eigenvectors.

The eigenvectors are usually normalized, so the final eigenvectors are

$$\boldsymbol{v}_1 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix} \quad \boldsymbol{v}_2 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}, \quad \text{and} \quad \boldsymbol{v}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

1.13 Spectral decomposition

In spectral decomposition, a second-order symmetric is expressed as a linear combination of its eigenvectors and eigenvalues. Given a second-order symmetric tensor $\boldsymbol{A}$ with eigenvalues λ_1, λ_2, and λ_3 and corresponding eigenvectors $\boldsymbol{v}_1$, $\boldsymbol{v}_2$, and $\boldsymbol{v}_3$, the tensor can be represented as

$$\boldsymbol{A} = \lambda_1 \boldsymbol{v}_1 \otimes \boldsymbol{v}_1 + \lambda_2 \boldsymbol{v}_2 \otimes \boldsymbol{v}_2 + \lambda_3 \boldsymbol{v}_3 \otimes \boldsymbol{v}_3 \tag{1.98}$$

Alternatively, we can write this in terms of projection tensors $\boldsymbol{P}_i$, defined by each eigenvector. The spectral decomposition then becomes

$$\boldsymbol{A} = \lambda_1 \boldsymbol{P}_1 + \lambda_2 \boldsymbol{P}_2 + \lambda_3 \boldsymbol{P}_3 = \lambda_i \boldsymbol{P}_i \tag{1.99}$$

where each projection tensor (also called an eigenprojectors or eigentensors) and are given by

$$\boldsymbol{P}_i = \boldsymbol{v}_i \otimes \boldsymbol{v}_i \quad \text{(no summation on } i\text{)} \tag{1.100}$$

The spectral decomposition can also be written as the following sum

$$\boldsymbol{A} = \lambda_i \boldsymbol{P}_i \tag{1.101}$$

The projection tensors $\boldsymbol{P}_i$ are orthogonal and idempotent, which means they satisfy: Since the projector tensors are orthogonal, they satisfy

$$\boldsymbol{P}_i \cdot \boldsymbol{P}_j = \delta_{ij}\boldsymbol{P}_i \tag{1.102}$$

Thus, for example, $\boldsymbol{P}_1\boldsymbol{P}_1 = \boldsymbol{P}_1^2 = \boldsymbol{P}_1$ and $\boldsymbol{P}_1\boldsymbol{P}_2 = \delta_{12}\boldsymbol{P}_1 = \boldsymbol{0}$.

The spectral decomposition is particularly useful for calculating powers of a tensor. For example, the square of a tensor $\boldsymbol{A}$ can be expressed as

$$\boldsymbol{A}^2 = (\lambda_1\boldsymbol{P}_1 + \lambda_2\boldsymbol{P}_2 + \lambda_3\boldsymbol{P}_3)(\lambda_1\boldsymbol{P}_1 + \lambda_2\boldsymbol{P}_2 + \lambda_3\boldsymbol{P}_3) \tag{1.103}$$

Using the orthogonality property of the projectors, this simplifies to:

$$\boldsymbol{A}^2 = \lambda_1^2\boldsymbol{P}_1 + \lambda_2^2\boldsymbol{P}_2 + \lambda_3^2\boldsymbol{P}_3 \tag{1.104}$$

Using spectral decomposition, any power n of $\boldsymbol{A}$ is given by

$$\boldsymbol{A}^n = \lambda_1^n\boldsymbol{P}_1 + \lambda_2^n\boldsymbol{P}_2 + \lambda_3^n\boldsymbol{P}_3 \tag{1.105}$$

This extends to fractional powers. For example, the square root of $\boldsymbol{A}$ is

$$\sqrt{\boldsymbol{A}} = \sqrt{\lambda_1}\boldsymbol{P}_1 + \sqrt{\lambda_2}\boldsymbol{P}_2 + \sqrt{\lambda_3}\boldsymbol{P}_3 \tag{1.106}$$

Example 1.17: Compute the square root of tensor $\boldsymbol{A}$

$$\boldsymbol{A} = \begin{pmatrix} 6.4 & 7.2 & 0 \\ 7.2 & 10.6 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

Solution:

Computing the eigenvalues and eigenvectors of $\boldsymbol{A}$, we find

$$\lambda_1 = 16, \quad \lambda_2 = 1, \quad \lambda_3 = 1$$

and

$$\boldsymbol{v}_1 = \begin{pmatrix} 0.6 \\ 0.8 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_2 = \begin{pmatrix} 0.8 \\ -0.6 \\ 0 \end{pmatrix}, \quad \boldsymbol{v}_3 = \begin{pmatrix} 0 \\ 0 \\ 1 \end{pmatrix}$$

respectively.

The square root of the tensor is then

$$\begin{aligned}
\sqrt{\boldsymbol{A}} &= \sqrt{16}\boldsymbol{P}_1 + \sqrt{1}\boldsymbol{P}_2 + \sqrt{1}\boldsymbol{P}_3 \\
&= 4\begin{pmatrix} 0.36 & 0.48 & 0 \\ 0.48 & 0.64 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0.64 & -0.48 & 0 \\ -0.48 & 0.36 & 0 \\ 0 & 0 & 0 \end{pmatrix} + \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \\
&= \begin{pmatrix} 2.08 & 1.44 & 0 \\ 1.44 & 2.92 & 0 \\ 0 & 0 & 1 \end{pmatrix}
\end{aligned}$$

1.14 Fundamental invariants

There are certain quantities associated to a tensor that remain invariant under coordinate transformations. These quantities are called invariants. For example, the length of a vector and the determinant of a second-order tensor are invariants since they are the same in all coordinate systems.

For a second-order tensor in 3D space, the fundamental invariants, also known as principal or characteristic invariants, are derived from its characteristic equation and are often expressed in terms of the tensor's eigenvalues or its components. These invariants are particularly useful in continuum mechanics, material science, and physics, where they help describe properties such as stress and strain in a material that do not change under orthogonal transformations (such as rotations).

The principal invariants of a second-order tensor $\boldsymbol{A}$ are defined as follows:

First invariant $\boldsymbol{I_1}$: The first invariant is the sum of the eigenvalues which is equivalent to the trace of the tensor

$$\begin{aligned} I_1 &= \lambda_1 + \lambda_2 + \lambda_3 \\ &= \operatorname{trace}(\boldsymbol{A}) \\ &= A_{ii} \\ &= A_{11} + A_{22} + A_{33} \end{aligned} \tag{1.107}$$

The first invariant is a measure of the overall magnitude of the tensor.

Second invariant $\boldsymbol{I_2}$: The second invariant I_2 is related to the sum of the principal minors of the tensor. It can also be expressed in terms of the trace and the square of the tensor. For a symmetric tensor, it is defined as:

$$\begin{aligned} I_2 &= \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 \\ &= \frac{1}{2}(\operatorname{trace}(\boldsymbol{A})^2 - \operatorname{trace}(\boldsymbol{A^2})) \\ &= \frac{1}{2}(I_1^2 - \operatorname{trace}(\boldsymbol{A^2})) \\ &= \frac{1}{2}(A_{ii}^2 - A_{ij}A_{ji}) \\ &= A_{11}A_{22} + A_{22}A_{33} + A_{33}A_{11} - A_{12}^2 - A_{23}^2 - A_{31}^2 \end{aligned} \tag{1.108}$$

This invariant can be interpreted as a measure of the "interaction" between the tensor's components.

Third invariant $\boldsymbol{I_3}$: The third invariant I_3 is the determinant of the tensor and is equal to the product of the eigenvalues

$$\begin{aligned} I_3 &= \det(\boldsymbol{A}) \\ &= \lambda_1\lambda_2\lambda_3 \\ &= A_{11}(A_{22}A_{33} - A_{23}A_{32}) - A_{12}(A_{21}A_{33} - A_{23}A_{31}) + A_{13}(A_{21}A_{23} - A_{22}A_{31}) \end{aligned} \tag{1.109}$$

In mechanics, this invariant is related to the volumetric change or the change in shape of a material.

Example 1.18: Given a second-order tensor $\boldsymbol{A}$ with components

$$\boldsymbol{A} = \begin{pmatrix} 1.64 & 0.48 & 0 \\ 0.48 & 1.36 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

calculate the principal invariants I_1, I_2, and I_3.

Solution:

For this tensor, expanding the characteristic equation $\det(\boldsymbol{A} - \lambda \boldsymbol{I}) = 0$ provides the following cubic polynomial

$$\lambda^3 - 4\lambda^2 + 5\lambda - 2 = 0$$

The roots of this equation are $\lambda_1 = 2$, $\lambda_2 = 1$, and $\lambda_3 = 1$. Then, the principal invariants are given by

$$\begin{aligned} I_1 &= \lambda_1 + \lambda_2 + \lambda_3 = 4 \\ I_2 &= \lambda_1\lambda_2 + \lambda_2\lambda_3 + \lambda_3\lambda_1 = 5 \\ I_3 &= \lambda_1\lambda_2\lambda_3 = 2 \end{aligned}$$

Example 1.19: Given a second-order symmetric tensor $\boldsymbol{A}$ calculate $\frac{\partial I_1}{\partial \boldsymbol{A}}$ and $\frac{\partial I_2}{\partial \boldsymbol{A}}$.

Solution:

For the derivative of the first invariant we have

$$\begin{aligned} \frac{\partial A_{kk}}{\partial A_{ij}} &= \frac{\partial A_{kn}\delta_{nk}}{\partial A_{ij}} \\ &= \delta_{ki}\delta_{nj}\delta_{nk} \\ &= \delta_{ij} \end{aligned}$$

Thus, $\frac{\partial I_1}{\partial \boldsymbol{A}} = \boldsymbol{I}$.

Regarding the derivative of the second invariant we have

$$\begin{aligned} \frac{\partial I_2}{\partial A_{ij}} &= \frac{\partial}{\partial A_{ij}}\left(\frac{1}{2}(A_{kk}^2 - A_{nm}A_{mn})\right) \\ &= \frac{1}{2}\left(2A_{kk}\delta_{ij} - 2A_{nm}\delta_{ni}\delta_{mj}\right) \\ &= \frac{1}{2}\left(2A_{kk}\delta_{ij} - 2A_{ij}\right) \\ &= A_{kk}\delta_{ij} - A_{ij} \end{aligned}$$

Then, $\frac{\partial I_2}{\partial \boldsymbol{A}} = \mathrm{trace}(\boldsymbol{A})\boldsymbol{I} - \boldsymbol{A}$.

Exercises 1.7:

1. Given the second-order tensor $\boldsymbol{A}$

$$\boldsymbol{A} = \begin{pmatrix} -4.216 & 0 & -2.688 \\ 0 & 12.5 & 0 \\ -2.688 & 0 & 4.216 \end{pmatrix}$$

calculate

a) The eigenvalues and eigenvectors.
b) The principal invariants I_1, I_2, and I_3.

1.15 Alternatively tensor representations

In tensor algebra, symmetric tensors can be efficiently represented by collecting only unique components. For example, second-order tensors can be compressed into 6-component vectors, and fourth-order tensors into 6×6 matrices. These simplified forms are frequently used in computational mechanics to optimize storage and facilitate tensor operations.

1.15.1 Voigt notation

In this notation, a second-order tensor $\boldsymbol{A}$ is transformed into a 6-component vector $\boldsymbol{A}_V$ by arranging its components as follows

$$\boldsymbol{A} = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{A}_V = \begin{pmatrix} A_{11} \\ A_{22} \\ A_{33} \\ A_{23} \\ A_{31} \\ A_{12} \end{pmatrix} \tag{1.110}$$

This vectorized form simplifies tensor operations computationally by treating tensors as vectors, but it does not preserve all tensor transformation properties.

1.15.2 Mandel notation

Mandel notation improves upon Voigt notation by introducing a scaling factor of $\sqrt{2}$ for shear components. This scaling accounts for the double appearance of shear terms in symmetric tensors, ensuring consistent norms and inner products. For a second-order tensor $\boldsymbol{A}$, the Mandel vector $\boldsymbol{A}_M$ is defined as

$$\boldsymbol{A} = \begin{pmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \\ A_{31} & A_{32} & A_{33} \end{pmatrix} \quad \Rightarrow \quad \boldsymbol{A}_M = \begin{pmatrix} A_{11} \\ A_{22} \\ A_{33} \\ \sqrt{2}A_{23} \\ \sqrt{2}A_{31} \\ \sqrt{2}A_{12} \end{pmatrix} \tag{1.111}$$

This ensures that

$$\|\boldsymbol{A}\| = \|\boldsymbol{A}_M\| \quad \text{and} \quad \boldsymbol{A}:\boldsymbol{B} = \boldsymbol{A}_M \cdot \boldsymbol{B}_M \tag{1.112}$$

For fourth-order tensors, Mandel notation converts an 81-component symmetric tensor into a 6×6 matrix with appropriate scaling for the shear terms. For example, the Mandel form of a fourth-order tensor $\boldsymbol{A}$ is

$$\boldsymbol{A}_M = \begin{pmatrix} A_{1111} & A_{1122} & A_{1133} & \sqrt{2}A_{1123} & \sqrt{2}A_{1113} & \sqrt{2}A_{1112} \\ A_{2211} & A_{2222} & A_{2233} & \sqrt{2}A_{2223} & \sqrt{2}A_{2213} & \sqrt{2}A_{2212} \\ A_{3311} & A_{3322} & A_{3333} & \sqrt{2}A_{3323} & \sqrt{2}A_{3313} & \sqrt{2}A_{3312} \\ \sqrt{2}A_{2311} & \sqrt{2}A_{2322} & \sqrt{2}A_{2333} & 2A_{2323} & 2A_{2313} & 2A_{2312} \\ \sqrt{2}A_{1311} & \sqrt{2}A_{1322} & \sqrt{2}A_{1333} & 2A_{1323} & 2A_{1313} & 2A_{1312} \\ \sqrt{2}A_{1211} & \sqrt{2}A_{1222} & \sqrt{2}A_{1233} & 2A_{1223} & 2A_{1213} & 2A_{1212} \end{pmatrix} \tag{1.113}$$

This matrix preserves consistency of contractions with second-order tensors. For example, for a fourth-order tensor $\boldsymbol{D}$ and a second-order tensor $\boldsymbol{\varepsilon}$, the double contraction in Mandel notation is

$$\boldsymbol{D}:\boldsymbol{\varepsilon} = \boldsymbol{D}_M \cdot \boldsymbol{\varepsilon}_M \tag{1.114}$$

where $\boldsymbol{D}_M$ is a 6×6 matrix and $\boldsymbol{\varepsilon}_M$ is a 6-component vector.

1.16 Curvilinear coordinate systems

In tensor algebra, curvilinear coordinates are a powerful way to describe geometries that are naturally non-rectilinear, making the calculations simpler or more intuitive. Common examples of curvilinear coordinates include polar, cylindrical and spherical coordinates. Figure 1.2 shows the the definitions of the cylindrical and spherical coordinates including the corresponding unit vectors.

1.16.1 Cylindrical coordinates

Cylindrical coordinates use three parameters to describe a point in space: the radial distance r, the azimuthal angle θ, and the axial distance z. The transformation between cylindrical and Cartesian coordinates is given by the following equations

$$\begin{aligned} x_1 &= r\cos(\varphi) \\ x_2 &= r\sin(\varphi) \\ x_3 &= z \end{aligned} \tag{1.115}$$

and

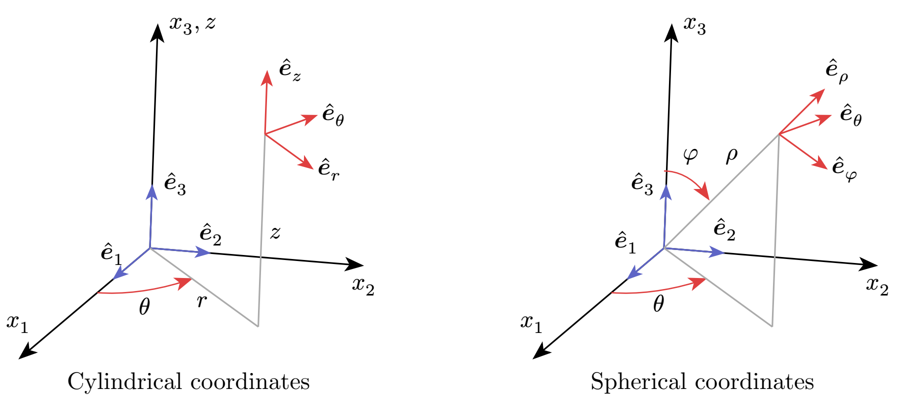

Figure 1.2: Curvilinear coordinates

$$
\begin{aligned}
r &= \sqrt{x_1^2 + y_1^2} \\
\theta &= \tan^{-1}\left(\frac{x_2}{x_1}\right) \\
z &= x_3
\end{aligned}
\tag{1.116}
$$

A vector $\boldsymbol{v}$ in cartesian coordinates can be written as

$$
\boldsymbol{v} = v_1\hat{\boldsymbol{e}}_1 + v_2\hat{\boldsymbol{e}}_2 + v_3\hat{\boldsymbol{e}}_3 \tag{1.117}
$$

while in cylindrical coordinates, it is expressed as

$$
\boldsymbol{v} = v_\rho\hat{\boldsymbol{e}}_\rho + v_\varphi\hat{\boldsymbol{e}}_\theta + v_z\hat{\boldsymbol{e}}_z \tag{1.118}
$$

where $\hat{\boldsymbol{e}}_\rho$, $\hat{\boldsymbol{e}}_\theta$, and $\hat{\boldsymbol{e}}_z$ are the unit vectors along the radial, azimuthal, and axial directions, respectively.

The gradient of a scalar function f in cylindrical coordinates can be derived as follows. First, let's express the ∇ operator in Cartesian coordinates:

$$
\nabla = \hat{\boldsymbol{e}}_1\frac{\partial}{\partial x_1} + \hat{\boldsymbol{e}}_2\frac{\partial}{\partial x_2} + \hat{\boldsymbol{e}}_3\frac{\partial}{\partial x_3} \tag{1.119}
$$

To convert to cylindrical coordinates, we need to express the unit vectors and derivates in terms of the cylindrical coordinates. The unit vectors in cylindrical coordinates are given by

$$
\begin{aligned}
\hat{\boldsymbol{e}}_r &= \cos(\theta)\hat{\boldsymbol{e}}_1 + \sin(\theta)\hat{\boldsymbol{e}}_2 \\
\hat{\boldsymbol{e}}_\theta &= -\sin(\theta)\hat{\boldsymbol{e}}_1 + \cos(\theta)\hat{\boldsymbol{e}}_2 \\
\hat{\boldsymbol{e}}_z &= \hat{\boldsymbol{e}}_3
\end{aligned}
\tag{1.120}
$$

Conversely

$$
\begin{aligned}
\hat{\boldsymbol{e}}_1 &= \cos(\theta)\hat{\boldsymbol{e}}_r - \sin(\theta)\hat{\boldsymbol{e}}_\theta \\
\hat{\boldsymbol{e}}_2 &= \sin(\theta)\hat{\boldsymbol{e}}_r + \cos(\theta)\hat{\boldsymbol{e}}_\theta \\
\hat{\boldsymbol{e}}_3 &= \hat{\boldsymbol{e}}_z
\end{aligned}
\tag{1.121}
$$

The derivatives can be expressed using the chain rule as

$$
\begin{aligned}
\frac{\partial}{\partial x_1} &= \frac{\partial}{\partial r}\frac{\partial r}{\partial x_1} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x_1} = \cos(\theta)\frac{\partial}{\partial r} + \left(-\frac{\sin(\theta)}{r}\right)\frac{\partial}{\partial \theta} \\
\frac{\partial}{\partial x_2} &= \frac{\partial}{\partial r}\frac{\partial r}{\partial x_2} + \frac{\partial}{\partial \theta}\frac{\partial \theta}{\partial x_2} = \sin(\theta)\frac{\partial}{\partial r} + \left(\frac{\cos(\theta)}{r}\right)\frac{\partial}{\partial \theta}
\end{aligned}
\tag{1.122}
$$

Subsituting into the ∇ operator we have:

$$
\begin{aligned}
\nabla &= \hat{\boldsymbol{e}}_1\frac{\partial}{\partial x_1} + \hat{\boldsymbol{e}}_2\frac{\partial}{\partial x_2} + \hat{\boldsymbol{e}}_3\frac{\partial}{\partial x_3} \\
&= (\cos(\theta)\hat{\boldsymbol{e}}_r - \sin(\theta)\hat{\boldsymbol{e}}_\theta)\left(\cos(\theta)\frac{\partial}{\partial r} + \left(-\frac{\sin(\theta)}{r}\right)\frac{\partial}{\partial \theta}\right) + \\
&\quad (\sin(\theta)\hat{\boldsymbol{e}}_r + \cos(\theta)\hat{\boldsymbol{e}}_\theta)\left(\sin(\theta)\frac{\partial}{\partial r} + \left(\frac{\cos(\theta)}{r}\right)\frac{\partial}{\partial \theta}\right) + \\
&\quad \hat{\boldsymbol{e}}_z\frac{\partial}{\partial z}
\end{aligned}
\tag{1.123}
$$

Simplifying we get

$$\nabla = \hat{\boldsymbol{e}}_r \frac{\partial}{\partial r} + \frac{1}{r}\hat{\boldsymbol{e}}_\theta \frac{\partial}{\partial \theta} + \hat{\boldsymbol{e}}_z \frac{\partial}{\partial z} \tag{1.124}$$

Thus, the gradient of a scalar function f in cylindrical coordinates is given by

$$\nabla f = \frac{\partial f}{\partial r}\hat{\boldsymbol{e}}_r + \frac{1}{r}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{e}}_\theta + \frac{\partial f}{\partial z}\hat{\boldsymbol{e}}_z \tag{1.125}$$

The divergence in cylindrical coordinates is given by

$$\begin{aligned} \nabla \cdot \boldsymbol{f} = &\left(\hat{\boldsymbol{e}}_r \frac{\partial}{\partial r} + \frac{1}{r}\hat{\boldsymbol{e}}_\theta \frac{\partial}{\partial \theta} + \hat{\boldsymbol{e}}_z \frac{\partial}{\partial z}\right) \cdot \left(f_\rho \hat{\boldsymbol{e}}_r + f_\theta \hat{\boldsymbol{e}}_\theta + f_z \hat{\boldsymbol{e}}_z\right) \\ = &\hat{\boldsymbol{e}}_r \cdot \frac{\partial}{\partial r}\left(f_\rho \hat{\boldsymbol{e}}_r + f_\theta \hat{\boldsymbol{e}}_\theta + f_z \hat{\boldsymbol{e}}_z\right) + \\ &\frac{1}{r}\hat{\boldsymbol{e}}_\theta \cdot \frac{\partial}{\partial \theta}\left(f_\rho \hat{\boldsymbol{e}}_r + f_\theta \hat{\boldsymbol{e}}_\theta + f_z \hat{\boldsymbol{e}}_z\right) + \\ &\hat{\boldsymbol{e}}_z \cdot \frac{\partial}{\partial z}\left(f_\rho \hat{\boldsymbol{e}}_r + f_\theta \hat{\boldsymbol{e}}_\theta + f_z \hat{\boldsymbol{e}}_z\right) + \end{aligned} \tag{1.126}$$

After deriving and simplifying the expression, we get the divergence in cylindrical coordinates as

$$\nabla \cdot \boldsymbol{f} = \frac{1}{r}\frac{\partial}{\partial r}(r f_r) + \frac{1}{r}\frac{\partial f_\theta}{\partial \theta} + \frac{\partial f_z}{\partial z} \tag{1.127}$$

In the above derivation, the following derivatives were used

$$\begin{aligned} &\frac{\partial \hat{\boldsymbol{e}}_r}{\partial r} = \mathbf{0}; \quad &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial r} = \mathbf{0}; \quad &&\frac{\partial \hat{\boldsymbol{e}}_z}{\partial r} = \mathbf{0}; \\ &\frac{\partial \hat{\boldsymbol{e}}_r}{\partial \theta} = \hat{\boldsymbol{e}}_\theta; \quad &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial \theta} = -\hat{\boldsymbol{e}}_r; \quad &&\frac{\partial \hat{\boldsymbol{e}}_z}{\partial \theta} = \mathbf{0}; \\ &\frac{\partial \hat{\boldsymbol{e}}_r}{\partial z} = \mathbf{0}; \quad &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial z} = \mathbf{0}; \quad &&\frac{\partial \hat{\boldsymbol{e}}_z}{\partial z} = \mathbf{0} \end{aligned} \tag{1.128}$$

Unlike Cartesian unit vectors, cylindrical unit vectors are not constant, i.e. they depend on the coordinates. Then, the derivatives of all unit vectors are not zero.

1.16.2 Spherical coordinates

Spherical coordinates are another common curvilinear coordinate system that uses three parameters to describe a point in space: the radial distance ρ, the polar angle φ, and the azimuthal angle θ. The transformation between spherical and Cartesian coordinates is given by the following equations

$$\begin{aligned} x_1 &= \rho \sin(\varphi)\cos(\theta) \\ x_2 &= \rho \sin(\varphi)\sin(\theta) \\ x_3 &= \rho \cos(\varphi) \end{aligned} \tag{1.129}$$

and

$$\begin{aligned}\rho &= \sqrt{x_1^2 + x_2^2 + x_3^2} \\ \theta &= \tan^{-1}\left(\frac{x_2}{x_1}\right) \\ \varphi &= \cos^{-1}\left(\frac{x_3}{\rho}\right)\end{aligned} \tag{1.130}$$

The unit vector in spherical coordinates are given by

$$\begin{aligned}\hat{\boldsymbol{e}}_\rho &= \sin(\varphi)\cos(\theta)\hat{\boldsymbol{e}}_1 + \sin(\varphi)\sin(\theta)\hat{\boldsymbol{e}}_2 + \cos(\varphi)\hat{\boldsymbol{e}}_3 \\ \hat{\boldsymbol{e}}_\varphi &= \cos(\varphi)\cos(\theta)\hat{\boldsymbol{e}}_1 + \cos(\varphi)\sin(\theta)\hat{\boldsymbol{e}}_2 - \sin(\varphi)\hat{\boldsymbol{e}}_3 \\ \hat{\boldsymbol{e}}_\theta &= -\sin(\theta)\hat{\boldsymbol{e}}_1 + \cos(\theta)\hat{\boldsymbol{e}}_2\end{aligned} \tag{1.131}$$

Conversely,

$$\begin{aligned}\hat{\boldsymbol{e}}_1 &= \sin(\varphi)\cos(\theta)\hat{\boldsymbol{e}}_\rho + \cos(\varphi)\cos(\theta)\hat{\boldsymbol{e}}_\varphi - \sin(\theta)\hat{\boldsymbol{e}}_\theta \\ \hat{\boldsymbol{e}}_2 &= \sin(\varphi)\sin(\theta)\hat{\boldsymbol{e}}_\rho + \cos(\varphi)\sin(\theta)\hat{\boldsymbol{e}}_\varphi + \cos(\theta)\hat{\boldsymbol{e}}_\theta \\ \hat{\boldsymbol{e}}_3 &= \cos(\varphi)\hat{\boldsymbol{e}}_\rho - \sin(\varphi)\hat{\boldsymbol{e}}_\varphi\end{aligned} \tag{1.132}$$

Following a similar procedeure as in the case of cylindrical coordinates, the ∇ operator in spherical coordinates is obtained as

$$\nabla = \hat{\boldsymbol{e}}_\rho \frac{\partial}{\partial \rho} + \frac{1}{\rho}\hat{\boldsymbol{e}}_\varphi \frac{\partial}{\partial \varphi} + \frac{1}{\rho\sin(\varphi)}\hat{\boldsymbol{e}}_\theta \frac{\partial}{\partial \theta} \tag{1.133}$$

Later, the gradient of a scalar function f in spherical coordinates can be expressed as

$$\nabla f = \frac{\partial f}{\partial \rho}\hat{\boldsymbol{e}}_\rho + \frac{1}{\rho}\frac{\partial f}{\partial \varphi}\hat{\boldsymbol{e}}_\varphi + \frac{1}{\rho\sin(\varphi)}\frac{\partial f}{\partial \theta}\hat{\boldsymbol{e}}_\theta \tag{1.134}$$

The divergence of a vector field $\boldsymbol{f}$ in spherical coordinates is given by

$$\nabla \cdot \boldsymbol{f} = \frac{1}{\rho^2}\frac{\partial}{\partial \rho}\left(\rho^2 f_\rho\right) + \frac{1}{\rho\sin(\varphi)}\frac{\partial}{\partial \varphi}\left(\sin(\varphi) f_\varphi\right) + \frac{1}{\rho\sin(\varphi)}\frac{\partial f_\theta}{\partial \theta} \tag{1.135}$$

Similarly to the case of cylindrical coordinates, the derivatives of the unit vectors in spherical coordinates are not zero, i.e.

$$\begin{aligned}&\frac{\partial \hat{\boldsymbol{e}}_\rho}{\partial \rho} = \mathbf{0}; &&\frac{\partial \hat{\boldsymbol{e}}_\varphi}{\partial \rho} = \mathbf{0}; &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial \rho} = \mathbf{0}; \\ &\frac{\partial \hat{\boldsymbol{e}}_\rho}{\partial \varphi} = \hat{\boldsymbol{e}}_\varphi; &&\frac{\partial \hat{\boldsymbol{e}}_\varphi}{\partial \varphi} = -\hat{\boldsymbol{e}}_\rho; &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial \varphi} = \mathbf{0}; \\ &\frac{\partial \hat{\boldsymbol{e}}_\rho}{\partial \theta} = \sin(\varphi)\hat{\boldsymbol{e}}_\theta; &&\frac{\partial \hat{\boldsymbol{e}}_\varphi}{\partial \theta} = \cos(\varphi)\hat{\boldsymbol{e}}_\theta; &&\frac{\partial \hat{\boldsymbol{e}}_\theta}{\partial \theta} = -\left(\sin(\varphi)\hat{\boldsymbol{e}}_\rho + \cos(\varphi)\hat{\boldsymbol{e}}_\varphi\right)\end{aligned} \tag{1.136}$$

Exercises 1.8:

1. Calculate $\nabla \cdot \boldsymbol{u}$ for the following vector fields assuming α and β are constants

   a) $u_r = u_\theta = 0, \quad u_z = \alpha + \beta r^2$
   b) $u_r = \frac{\sin(\theta)}{r} = 0, \quad u_\theta = u_z = 0$
   c) $u_r = r^2\sin(\theta) = 0, \quad u_\theta = r^2\cos(\theta) = 0, \quad u_z = 0$

2. Calculate $\nabla \cdot \boldsymbol{u}$ for the vector field given in spherical coordinates assuming α and β are constants

$$u_\rho = \alpha\rho + \frac{\beta}{\rho^2}, \quad u_\theta = u_\varphi = 0$$

3. Demonstrate that $\frac{\partial \hat{\boldsymbol{e}}_\rho}{\partial \theta} = \sin(\varphi)\hat{\boldsymbol{e}}_\theta$ in spherical coordinates.